fin301 / FinTech301 | Indoor Mapping

Camera-assisted planar LiDAR | Group: [enter group ID]

DRAFT - INTERNAL EVALUATION; PHYSICAL VALIDATION INCOMPLETE.
Estimated duration: 591.8 / 591.9 s (100.0% of raw-scan span).

Offline reconstruction uses the supplied physical-car ROS 2 bag: a monocular RGB camera and raw planar LiDAR. No IMU, GNSS, magnetometer, wheel/recorded odometry, recorded poses, dynamic TF or recorded map enters estimation. The supplied recording is used as received; this report makes no claim of new data collection.

Inputs and time alignment

17,628 raw scans (29.78 Hz); 5,879 RGB messages (10.00 Hz); 2 decode failures. Integer header stamps [1] associate RGB to LiDAR within 50 ms. Nearest-scan timing median/p95: 8.40/15.85 ms. RGB starts 3.033 s later; initial poses use LiDAR alone. Recorded intrinsics and static extrinsics are used; physical calibration is unverified.

Estimation and development validation

Robust point-to-line ICP registers raw LiDAR to local submaps. Camera appearance proposes loops; 0 pass visual/geometric checks. No camera pose correction is applied. 1 LiDAR revisit edge(s) correct the SE(2) graph [3]; maximum correction 1.498 m. Engineering issue: valid-normal support did not measure overlap. Separate coverage and reciprocal/neighbor checks handle partial overlap. Camera features are triangulated from first/last views using metric LiDAR poses and checked in all 6 construction views [2]. These sparse candidate landmarks are experimental. Held-out raw scans use eight starts (0.25/0.50 m, 5/10 deg), with training scans excluded within 5 s or 0.5 m. Stable means all eight fits pass and agree within 0.15 m / 3 deg. Spread uses 92 queries with at least two accepted starts; failed starts count against acceptance. Fixed baseline seeds and reference IDs support internal comparison, not ground truth.

Coordinates and mounting

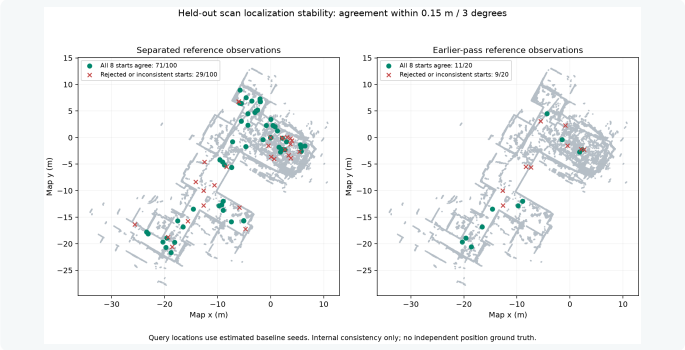
First accepted LiDAR pose defines local x forward, y left, metres; LiDAR z=0 is the scan plane. Recorded static mounting converts laser to vehicle poses when available. Visual heights assume planar motion. Full calibration and transforms remain in the audit.

Internal development validation

Tracked duration / bag span	**591.8 / 591.9 s**
LiDAR / camera candidate points	**49,783 / 29**
Accepted LiDAR / camera revisits	**1 / 0**
Held-out accepted starts	**712 / 800**
All starts agree / queries	**71 / 100**
Within-query spread p95: m / deg	**0.009 / 0.18**
Accepted fit median / p95 (mm)	**14.7 / 71.0**
Earlier-pass stable / queries	**11 / 20**

Export and inspection

map.pcd is PCD 0.7 ASCII XYZ in metres [4]: LiDAR scan-plane points (z=0) plus triangulated camera XYZ. Use lidar_scan_map.pcd or LiDAR occupancy for 2D localization; visual_landmarks.pcd is a separate experimental component. Do not project visual XYZ into a 2D localization map. The combined map is for source inspection, not validated 3D navigation. Open with PCL pcl_viewer or CloudCompare with PCD support.



Development validation: green means all eight initializations agree; red means a rejected or inconsistent start. Queries are held out from odometry, but used for method selection. Positions are estimated.



Camera candidate landmarks: 29 XYZ points after 6-view in-sample screening. Not a loop closure, independent height reference or demonstrated localization improvement.

Limits and missing experiment

Earlier four-view diagnostic: only 10/93 landmarks pass the 2 px prediction gate in both additional frames with frozen XYZ and trajectory. The selected 6-view method uses its construction views for in-sample screening; those views are not a new held-out test. Camera landmarks do not correct poses or establish localization improvement. Same internal test: stable queries 67 to 71/100; accepted starts 707 to 712/800. This test informed selection; generalization remains untested. Visual reprojection and ICP residuals are internal consistency measures, not independent 3D or pose accuracy. Planar-motion, recorded calibration, moving objects, low texture and uncompensated scan motion limit reliability.
Physical-reference validation remains incomplete; no reference provided. Measured walls, landmarks or an independent trajectory are needed; no centimetre-accuracy claim is supported.

References and reproducibility

[1] ROS 2 LaserScan (github.com/ros2/common_interfaces); [2] OpenCV calib3d (docs.opencv.org); [3] SciPy least_squares (docs.scipy.org); [4] PCL PCD format (pointclouds.org).
Evidence: outputs/fin301/audit/audit.json; outputs/fin301/result/metrics.json, loop_events.json, trajectory_arrays.npz, map.pcd. Evaluation JSON paths, reproduction command and editable text: group_technical_report.md; source hashes: report_manifest.json.